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Practical 3

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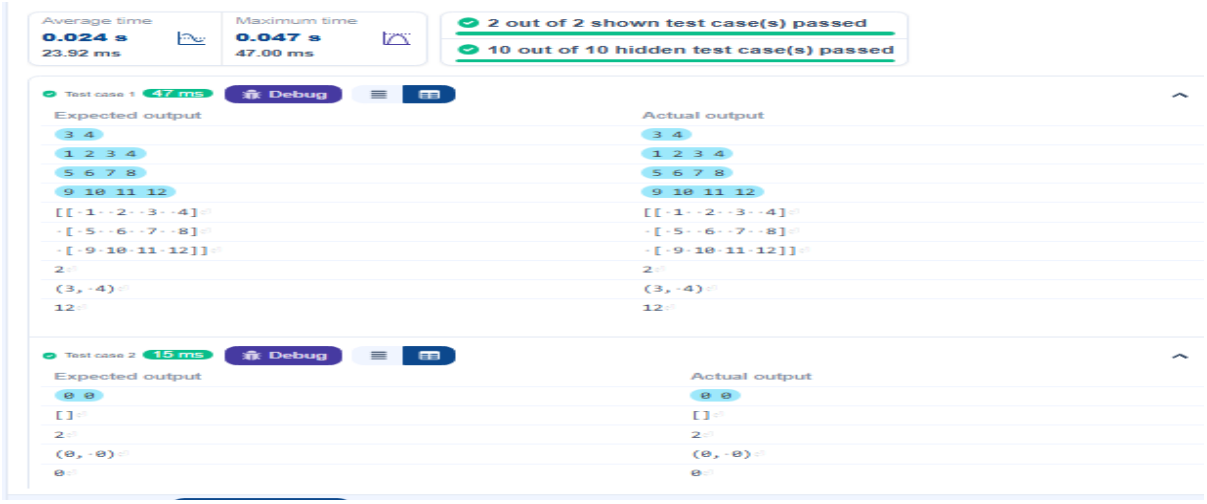
1)

Program :

```
import numpy as np

rows, cols = map(int, input().split())

elements = []

for _ in range(rows) :

        elements.extend(map(int, input().split()))

array = np.array(elements).reshape(rows, cols)

print(array)

print(array.ndim)

print(array.shape)

print(array.size)
```

Output :



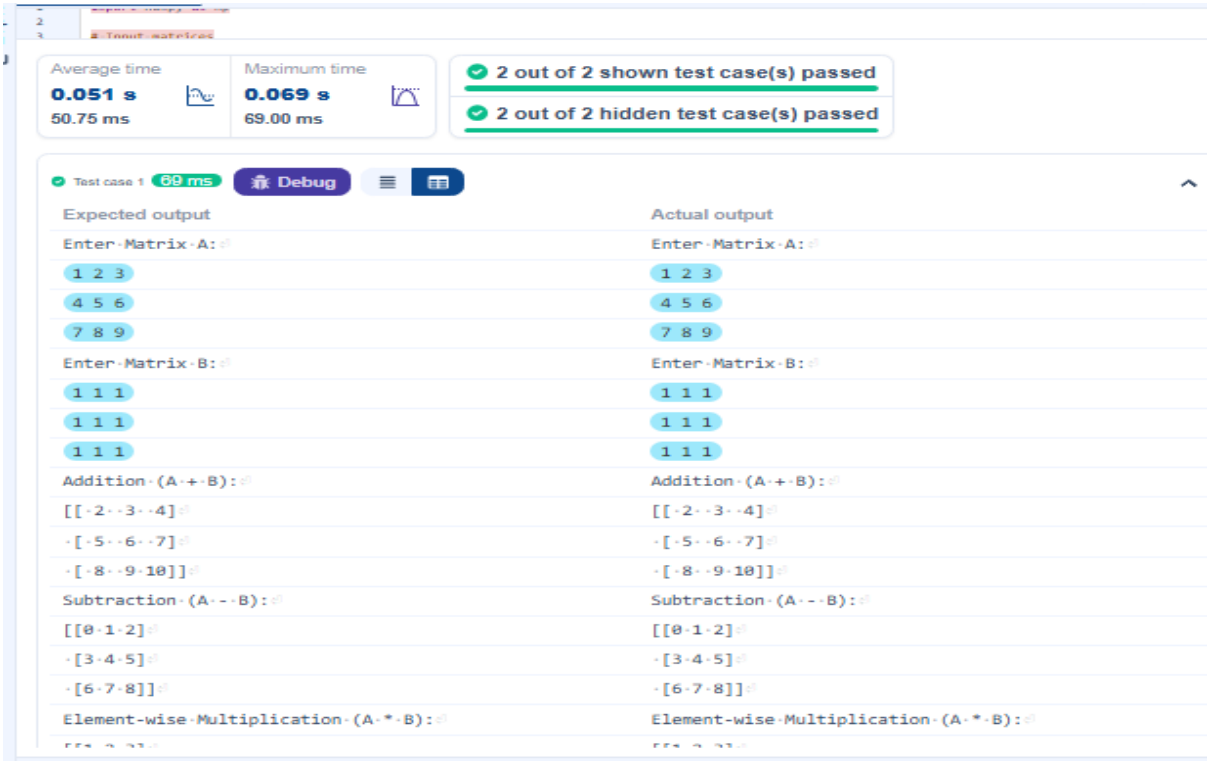
2)

Program :

```
import numpy as np

# Input matrices

print("Enter Matrix A:")

matrix_a = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Matrix B:")

matrix_b = np.array([list(map(int, input().split())) for i in range(3)])

# Addition

addition_result = matrix_a + matrix_b

print("Addition (A + B):")

print(addition_result)

# Subtraction

subtraction_result = matrix_a - matrix_b

print("Subtraction (A - B):")

print(subtraction_result)

# Multiplication (element-wise)

elementwise_multiplication_result = matrix_a * matrix_b

print("Element-wise Multiplication (A * B):")

print(elementwise_multiplication_result)

# Matrix multiplication (dot product)

matrix_multiplication_result = np.dot(matrix_a, matrix_b)

print("A dot B:")

print(matrix_multiplication_result)

# Transpose

transpose_a = matrix_a.T

print("Transpose of A:")
```

print(transpose_a)

Output :



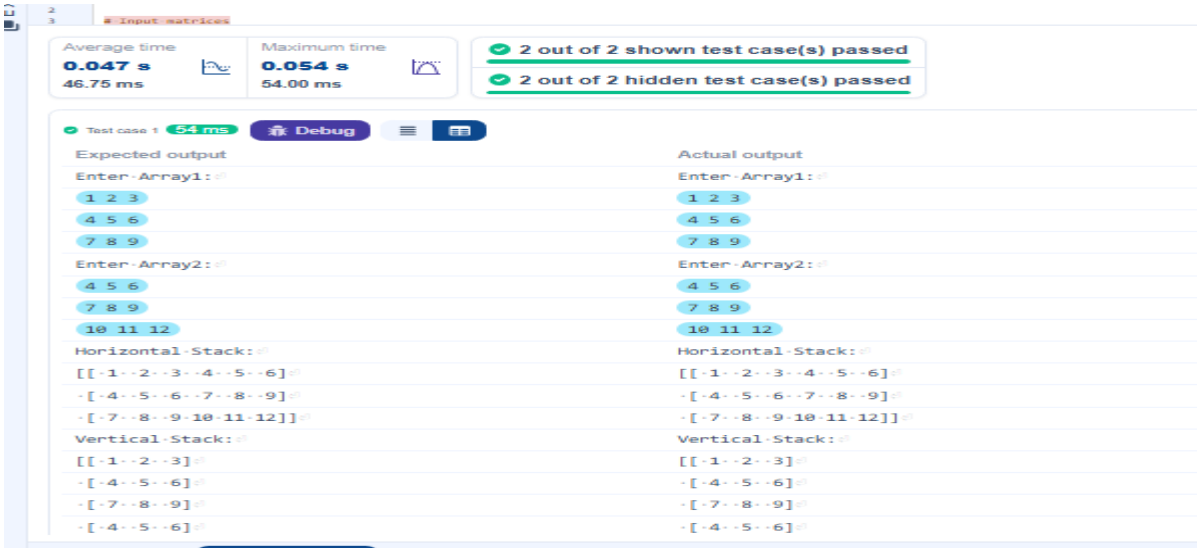
3)

Program :

```
import numpy as np

# Input matrices

print("Enter Array1:")

arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")

arr2 = np.array([list(map(int, input().split())) for i in range(3)])

# Perform horizontal stacking (hstack)

a = np.hstack((arr1,arr2))

print("Horizontal Stack:")

print(a)
```

```
# Perform vertical stacking (vstack)

b = np.vstack((arr1,arr2))

print("Vertical Stack:")

print(b)
```

Output :

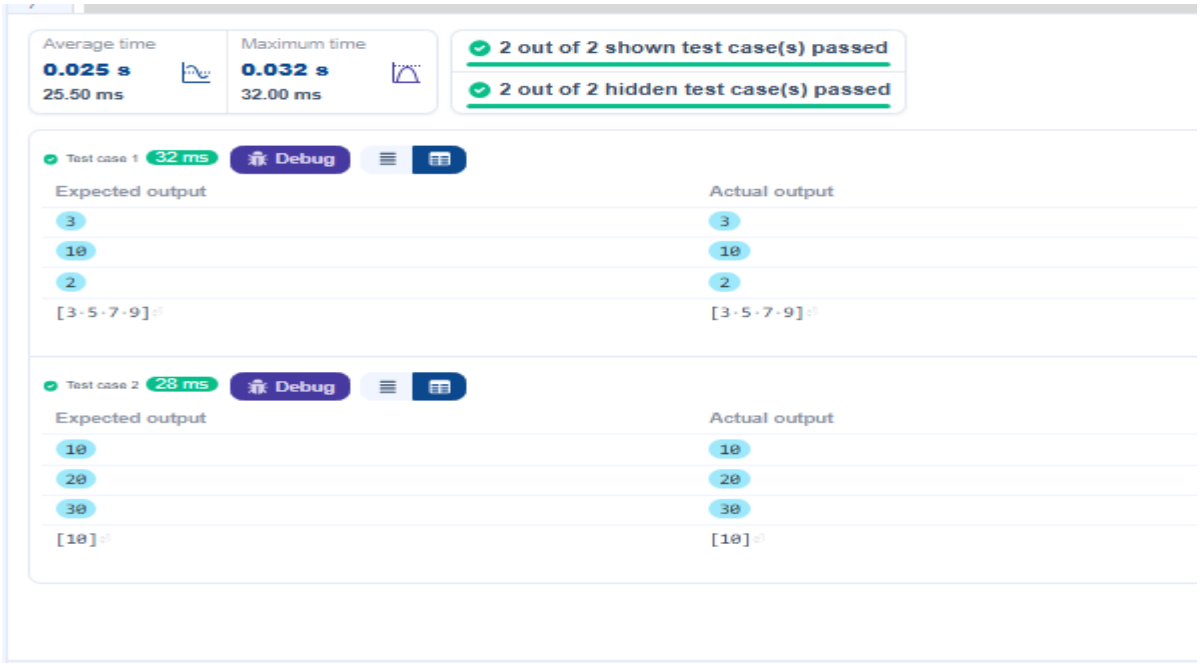


4)

Program :

```
import numpy as np

# Take user input for the start, stop, and step of the sequence

start = int(input())

stop = int(input())

step = int(input())

# Generate the sequence using np.arange()

a= np.arange(start,stop, step)

print(a)

# Print the generated sequence
```

Output :



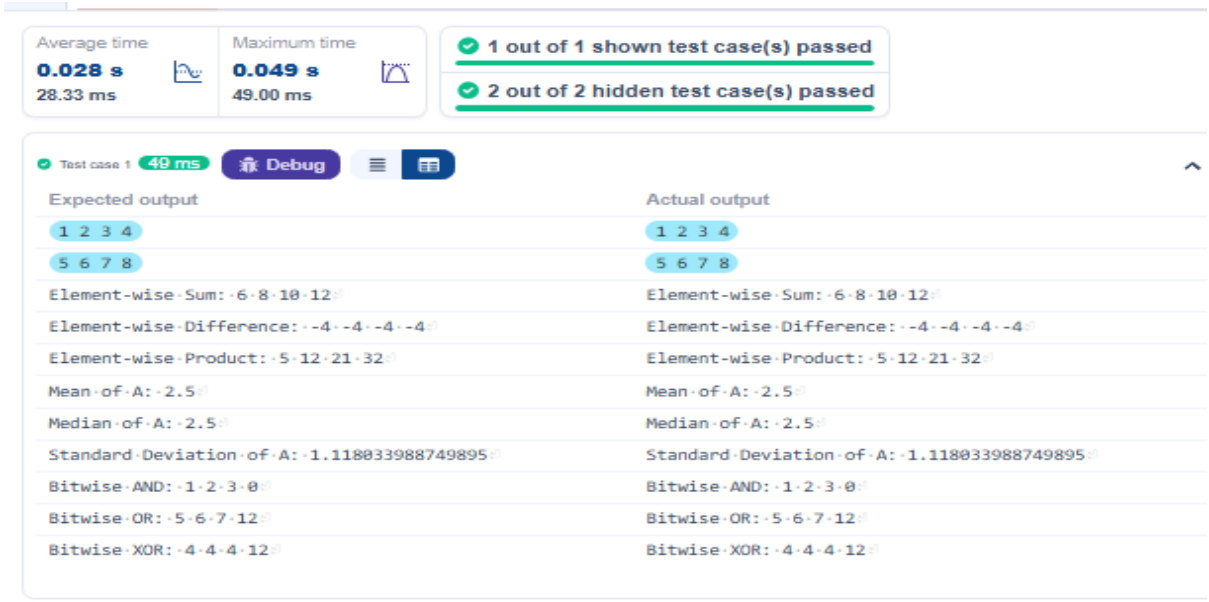
5)

Program :

```
import numpy as np


def array_operations(A, B):


    # Convert A and B to NumPy arrays

    A = np.array(A)

    B = np.array(B)


    # Arithmetic Operations

    sum_result = A + B

    diff_result = A - B

    prod_result = A * B
```

```
    # Statistical Operations
    mean_A = np.mean(A)
    median_A = np.median(A)
    std_dev_A = np.std(A)

    # Bitwise Operations
    and_result = A & B
    or_result = A | B
    xor_result = A ^ B

  # Output results with one space between each element
    print("Element-wise Sum:", ' '.join(map(str, sum_result)))
    print("Element-wise Difference:", ' '.join(map(str, diff_result)))
    print("Element-wise Product:", ' '.join(map(str, prod_result)))

    print(f"Mean of A: {mean_A}")
    print(f"Median of A: {median_A}")
    print(f"Standard Deviation of A: {std_dev_A}")

    print("Bitwise AND:", ' '.join(map(str, and_result)))
    print("Bitwise OR:", ' '.join(map(str, or_result)))
    print("Bitwise XOR:", ' '.join(map(str, xor_result)))

A = list(map(int, input().split()))  # Elements of array A
B = list(map(int, input().split()))  # Elements of array B
array_operations(A, B)
```

Output :



6)

Program :

```
import numpy as np


inputlist = list(map(int,input().split(" ")))


# Original array

original_array = np.array(inputlist)


# Create a view

view_array = original_array.view()


# Create a copy

copy_array = original_array.copy()
```

Modify the view

view_array[0] = 99

print("Original array after modifying view:", original_array)

print("View array:", view_array)

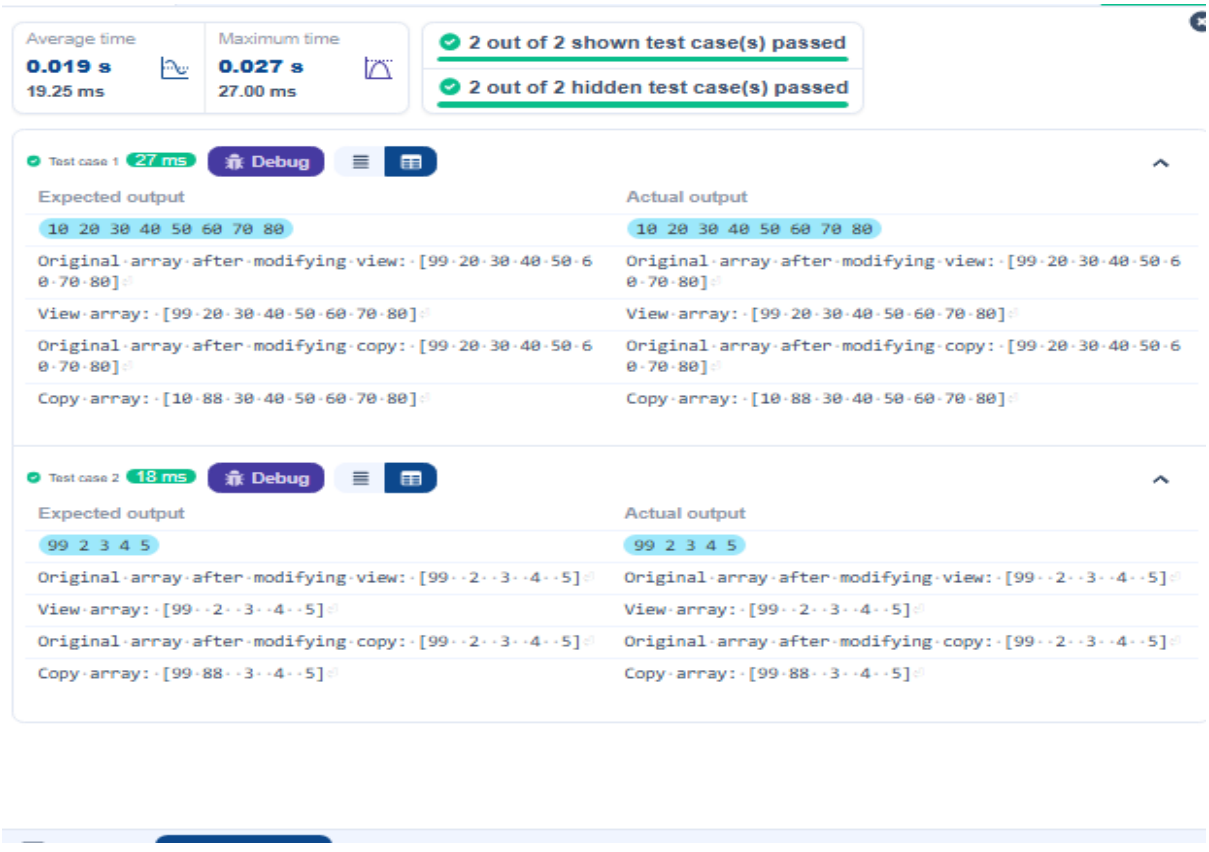
Modify the copy

copy_array[1] = 88

print("Original array after modifying copy:", original_array)

print("Copy array:", copy_array)

Output :

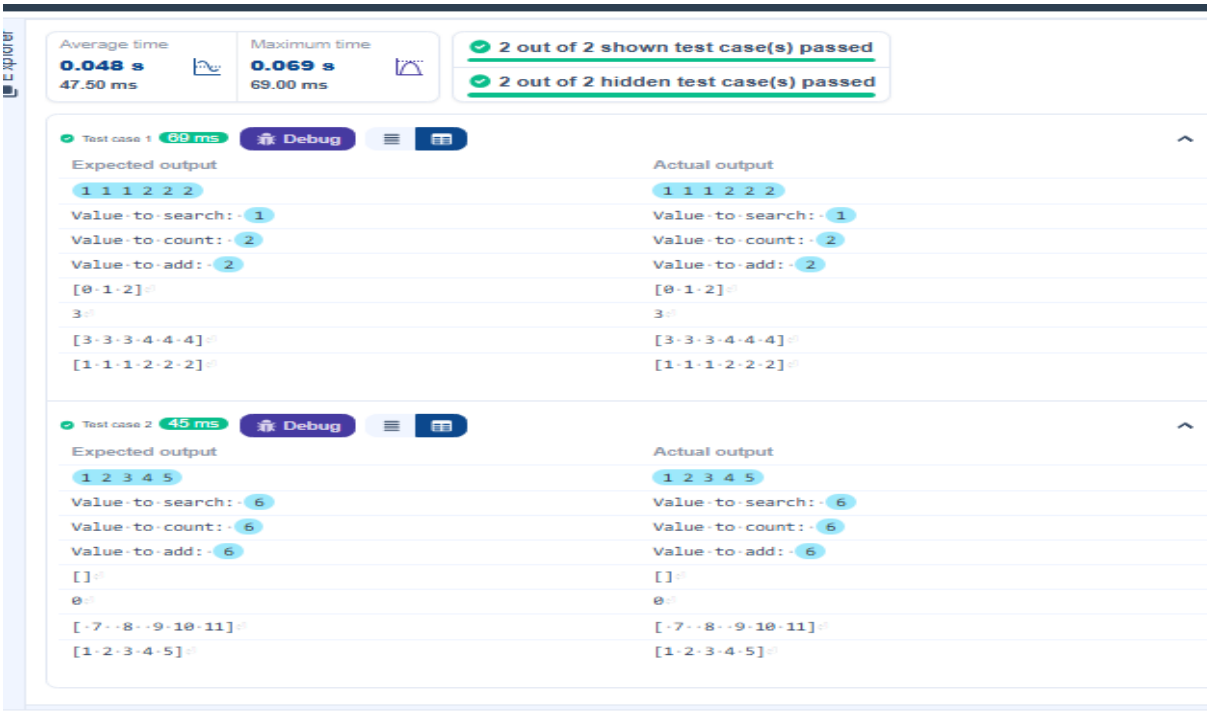


7)

Program :

```
import numpy as np


# Input array from the user

array1 = np.array(list(map(int, input().split())))


# Searching

search_value = int(input("Value to search: "))

count_value = int(input("Value to count: "))

broadcast_value = int(input("Value to add: "))


# Find indices where value matches in array1

a=np.where(array1==search_value)[0]

print(a)

# Count occurrences in array1

b=np.count_nonzero(array1==count_value)

print(b)

# Broadcasting addition

c=array1 + broadcast_value

print(c)

# Sort the first array

d=np.sort(array1)

print(d)
```

Output :



8)

Program :

```
import numpy as np

a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

print("All student Details:\n",a)

print("Total Students:", len(a))

print("All Student Roll Nos", a[:,0] )

print("Subject 1 Marks", a[:,1])

print("Min marks in Subject 2", a[:,2].min()     )

print("Max marks in Subject 3", a[:,3].max()    )

a1 = np.delete(a,0,1)

print("All subject marks:", a1      )

print("Total Marks", a[:,1]+a[:,2]+a[:,3]       )

a2 = np.array(a[:,1]+a[:,2]+a[:,3])
```

```
A =np.around(a2/3,1)
print(A)
s1 = (a[:,1]).mean()
s2 = (a[:,2]).mean()
s3 = (a[:,3]).mean()
A1 = np.array((s1,s2,s3))
print("Average Marks of each subject", A1      )
A2 = np.array((s1,s2))
print("Average Marks of S1 and S2",    A2      )
A3 = np.array((s1,s3))
print("Average Marks of S1 and S3",    A3           )
i = (a[:,3]).argmax()
r = a[i][0]
print("Roll no who got maximum marks in Subject 3",  r        )
I = (a[:,2]).argmin()
R = a[I][0]
print("Roll no who got minimum marks in Subject 2",  R    )
r1 = (a[(a[:,2])== 24 ,0]).reshape(-1,1)
print("Roll no who got 24 marks in Subject 2",     r1           )
count = np.argwhere(a[:,1]<40)
c = len(count)
print("Count of students who got marks in Subject 1 < 40",        c      )
count1 = np.argwhere(a[:,2]>90)
c1 = len(count1)
print("Count of students who got marks in Subject 2 > 90:",       c1      )
Count = [len(np.argwhere(a[:,1]>=90)),len(np.argwhere(a[:,2]>=90)),len(np.argwhere(a[:,3]>=90))]
Count = np.array(Count)
```

```
print("Count of students in each subject who got marks >= 90:",    Count         )
print("Roll no:",  a[:,0]    )
count0 = []
for i in range(len(a[:,0])):
        val = 0
        if a[i][1]>=90:
                val = val + 1
        elif a[i][2]>=90:
                val = val + 1
        elif a[i][3]>=90:
                val = val + 1
        count0.append(val)
count0 = np.array(count0)
print("Count of subjects in which student got marks >= 90:",    count0   )
asc = np.sort(a[:,1])
print(asc)
print(a[a[:,1]>=50])
print(a[a[:,1]<=90])
print(np.where(a[:,1]==79))
```

Output :


Average time
0.044 s
44.00 ms
Maximum time
0.044 s
44.00 ms
1 out of 1 shown test case(s) passed
Test case 1
44 ms
Debug
Expected output
Actual output
All student Details:
[[301. 67. 77. 88.]
[302. 78. 88. 77.]
[303. 45. 56. 89.]
[304. 88. 98. 45.]
[305. 78. 88. 99.]
[306. 68. 78. 36.]
[307. 79. 12. 45.]
[308. 46. 45. 77.]
[309. 89. 32. 88.]
[310. 79. 24. 33.]]
Total Students: 10
All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.]
Min marks in Subject 2 12.0
Max marks in Subject 3 99.0
All subject marks: [[67. 77. 88.]
[78. 88. 77.]